

# CS230: Lecture 2 Key AI Concepts Through Case Studies

Kian Katanforoosh

# Today's outline

- I. Recap' of the week
- II. Supervised Learning Projects
  - A. Day & Night Classification
  - B. Trigger Word Detection
  - C. Face Verification
- III. Self-Supervised Learning & Weakly Supervised Learning Projects
  - A. Image Embeddings
  - B. Multi-Modal Embeddings

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## I. **Recap' of the week**

## II. Supervised Learning Projects

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## III. Self-Supervised Learning & Weakly Supervised Learning Projects

- A. Image Embeddings
- B. Multi-Modal Embeddings

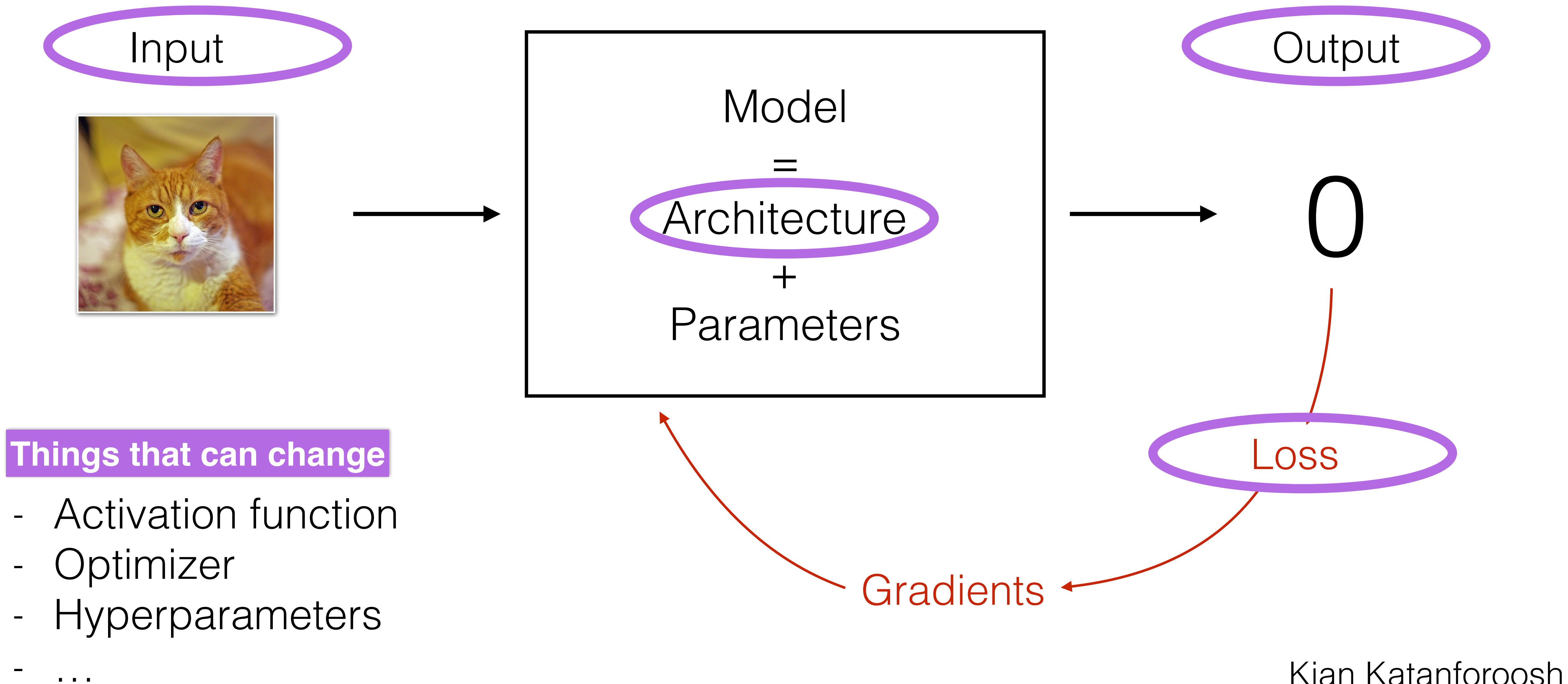




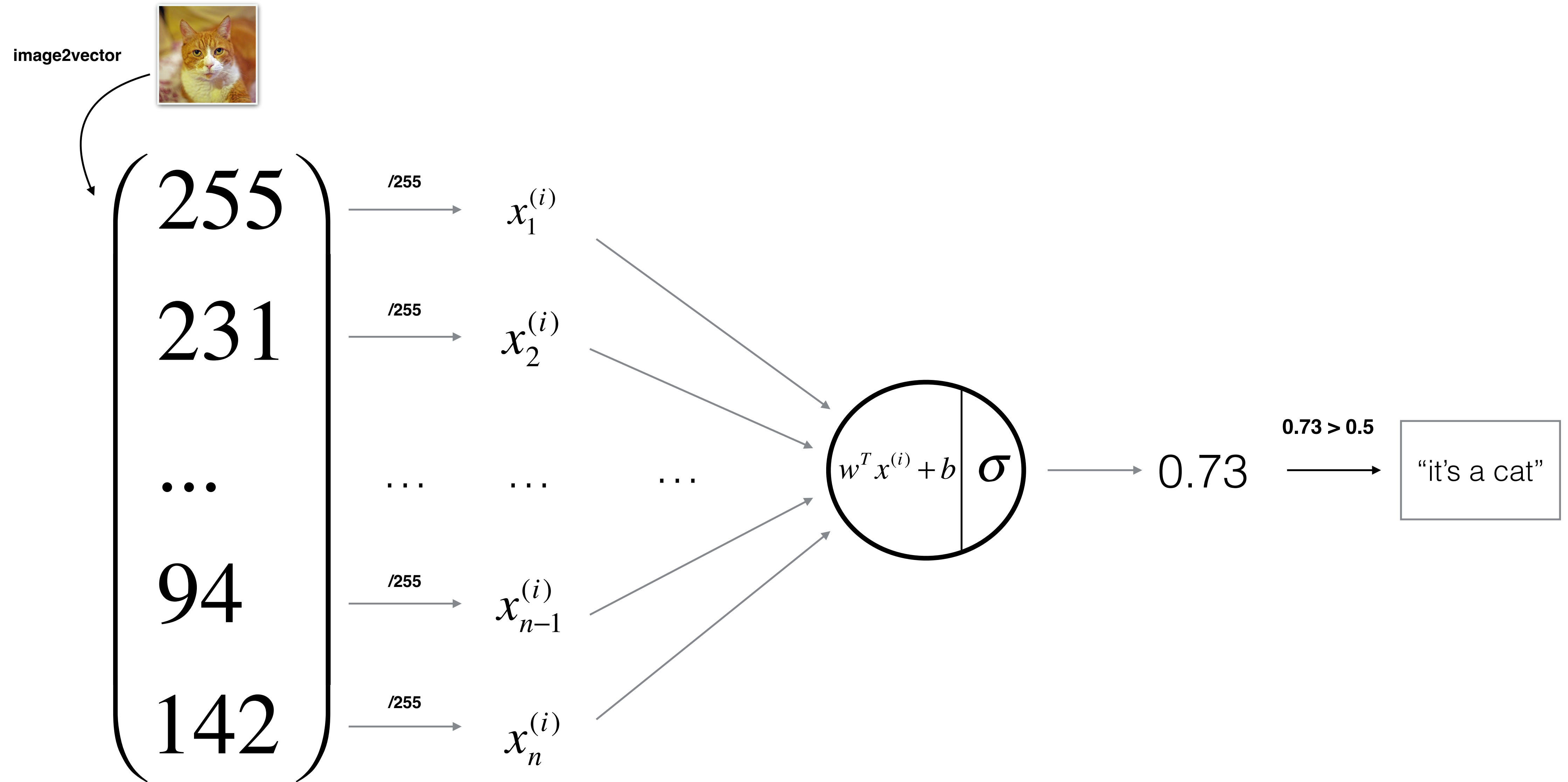
Recap of the week



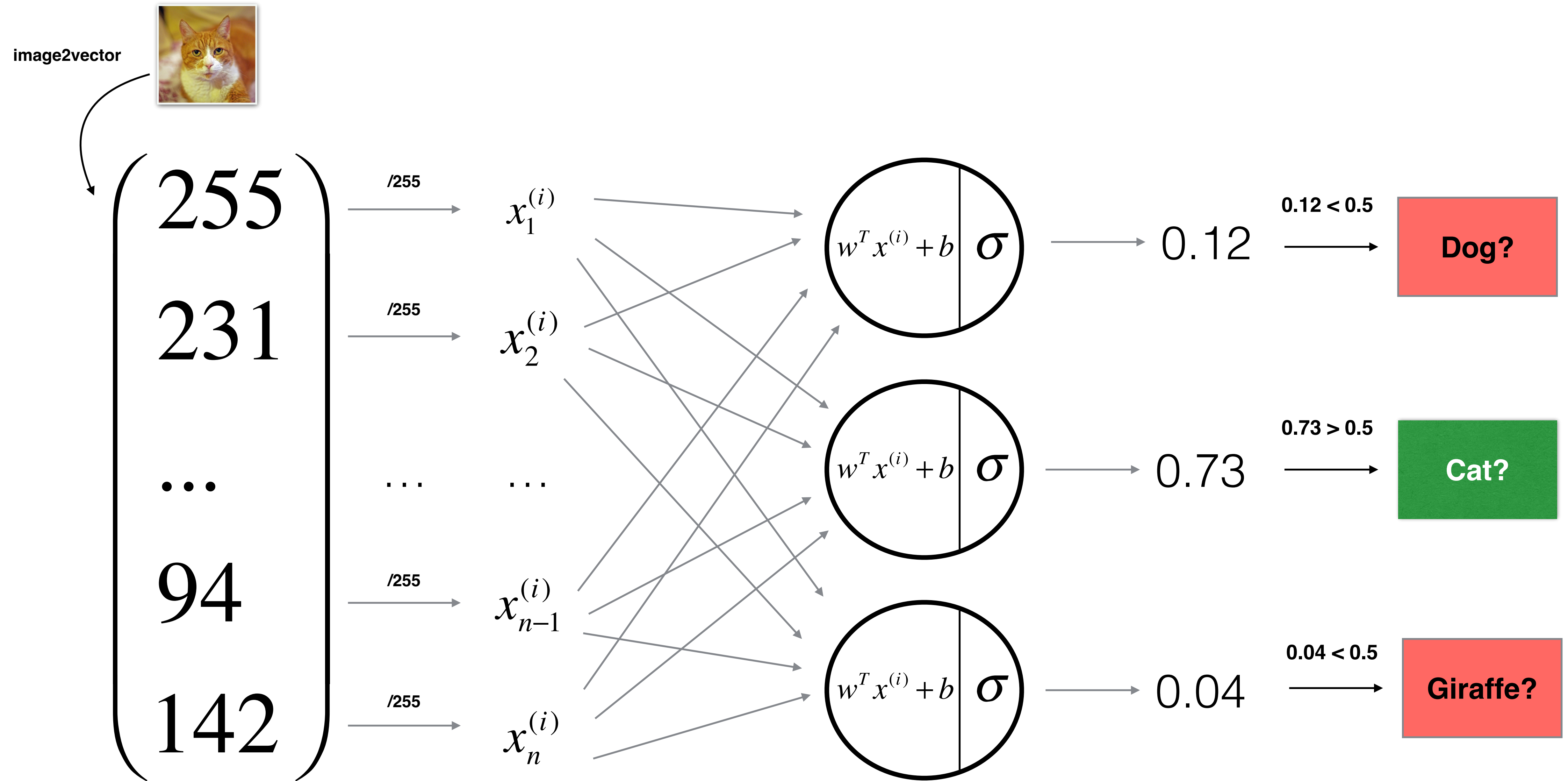
# Learning Process



# Logistic Regression as a Neural Network

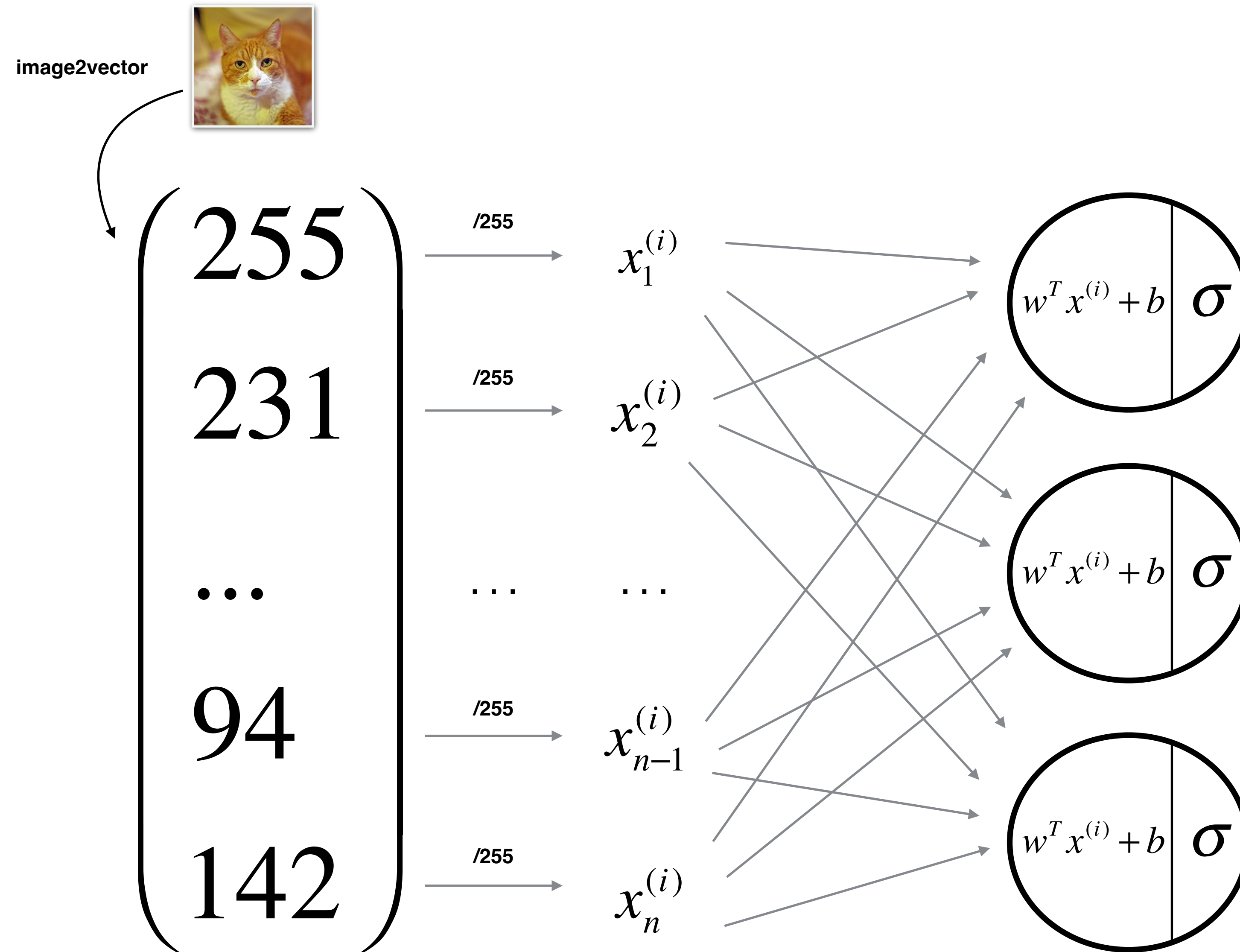


# Multi-class

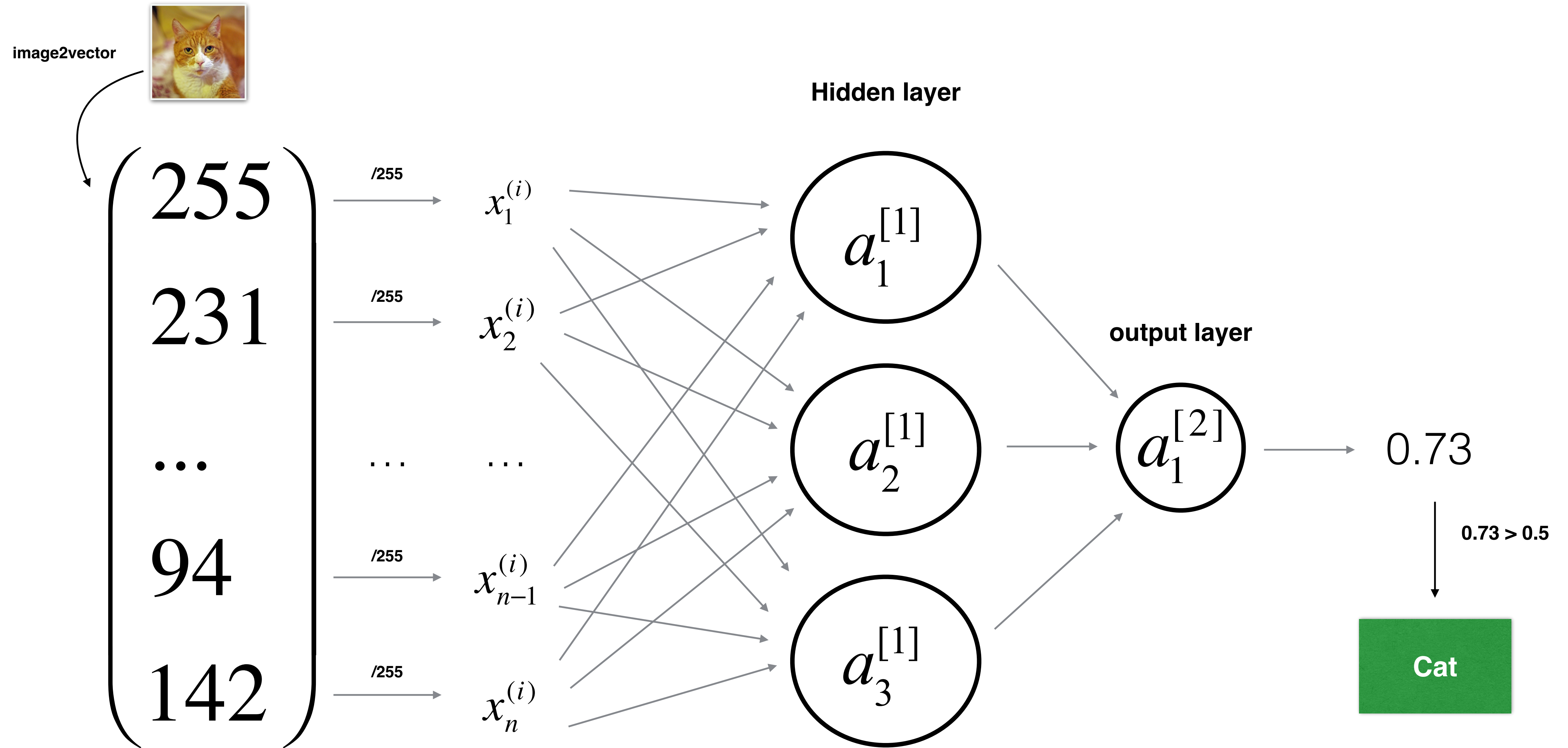




# Neural Network (Multi-class)

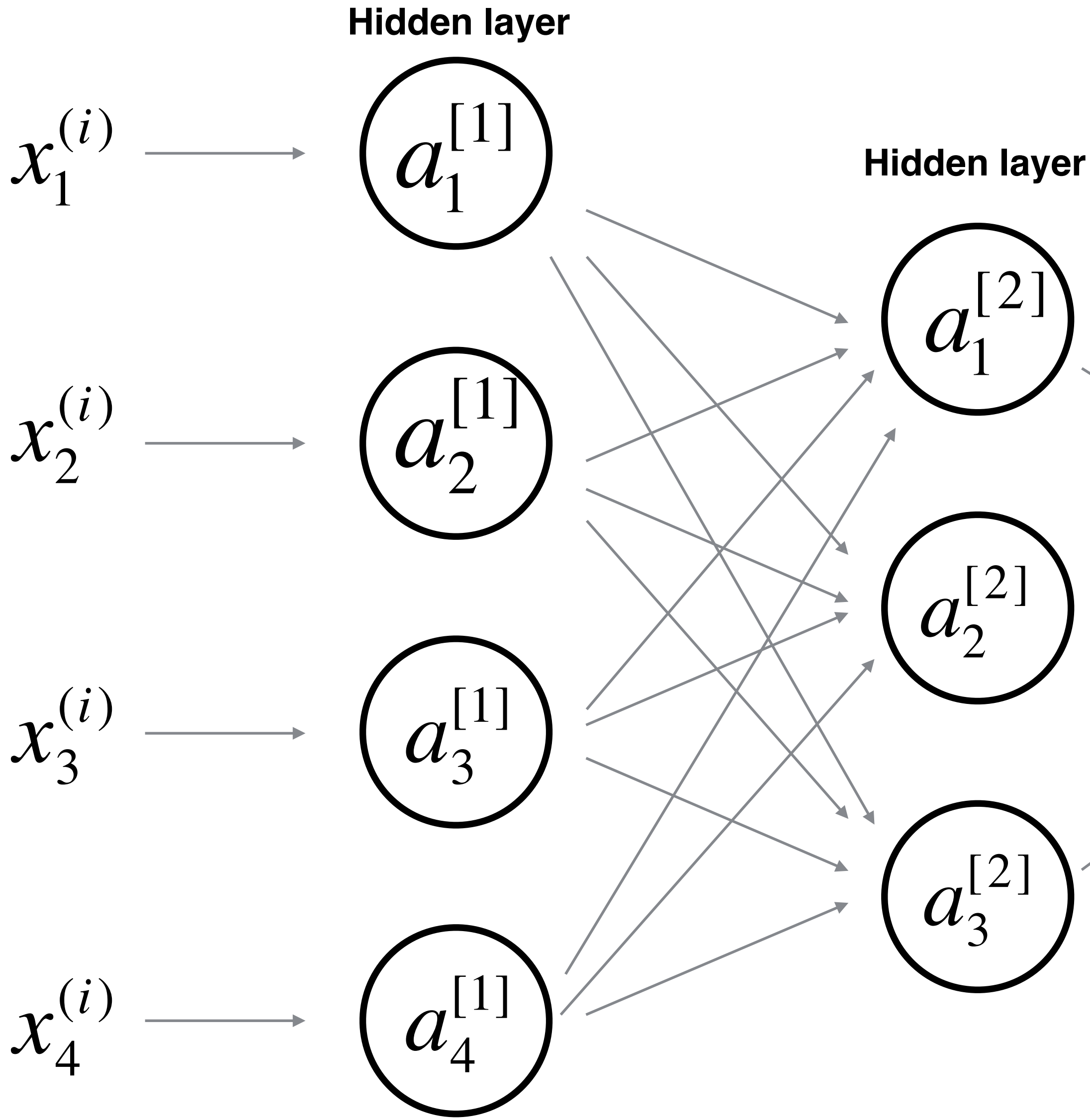


# Neural Network (1 hidden layer)





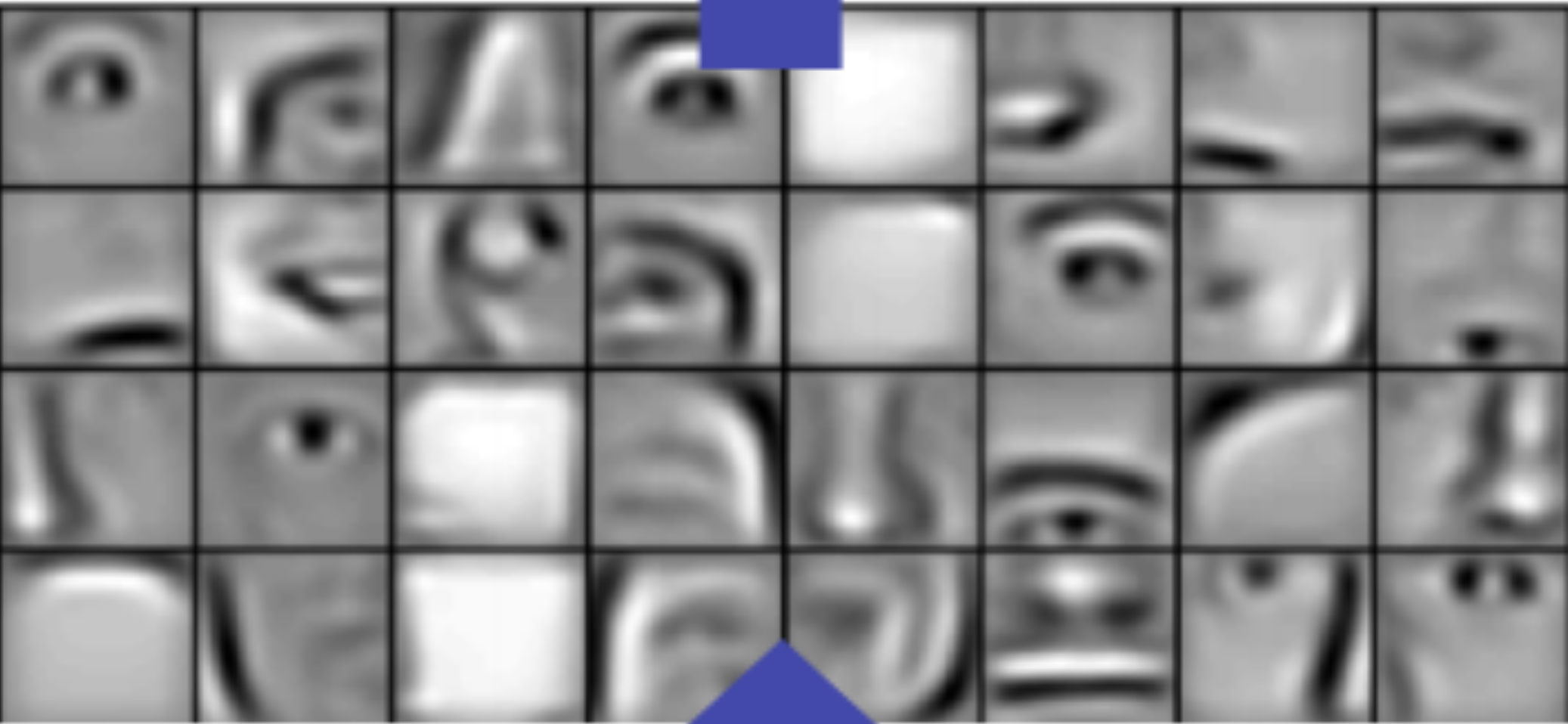
Deeper net



Technique called “encoding”



Layer 3



Layer 2



Layer 1

# Summary of learnings: Introduction

## New Words We Learned

- **Model = Architecture + Parameters**
  - **Architecture:** The blueprint or design of the neural network.
  - **Parameters:** The numerical values (weights, biases) the model learns during training.
- **Feature Engineering:** *Manual* — humans specify features.
- **Feature Learning:** *Automatic* — the network figures out features during training.
- **Encoding:** Any representation in vector form.
- **Embedding:** An encoding where **closeness = similarity**.
- **One-Hot Vector:** Exactly one thing is true.
- **Multi-Hot Vector:** Multiple things can be true at the same time.



~~Recap of the week~~



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# A) Day & Night Classification

**Goal:** Given an image, classify as taken “during the day” (0) or “during the night” (1)

## 1. Data?

10,000 images

Split? Bias?

## 2. Input?



Resolution?

(64, 64, 3)

## 3. Output?

$y = 0$  or  $y = 1$

Last Activation?

sigmoid

## 4. Architecture ?

A shallow CNN should do the job pretty well

## 5. Loss?

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

Easy warm up

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# Summary of learnings: Day 'n' Night classification

- Use a known **proxy project** to evaluate how much data you need.
- Be scrappy. For example, if you'd like to find a good resolution of images to use for your data, but don't have time for a large scale experiment, **approximate human-level performance by testing your friends** as classifiers.



# Case study: Trigger word detection

**Goal:** Given a 10sec audio speech, detect the word “activate”.

**1. Data?**

A bunch of 10s audio clips

Distribution?

**2. Input?**

$x =$  A 10sec audio clip

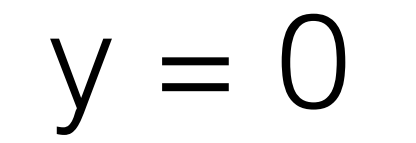


Resolution? (sample rate)

**3. Output?**

$y = 0$  or  $y = 1$

**Let's have an experiment!**



# Case study: Trigger word detection

**Goal:** Given a 10sec audio speech, detect the word “activate”.

## 1. Data?

A bunch of 10s audio clips

Distribution?

## 2. Input?

$x =$  A 10sec audio clip



Resolution? (sample rate)

## 3. Output?

$y = 0$  or  $y = 1$

$y = 00..0000\mathbf{1}00000..000$

$y = 00..0000\mathbf{1}..\mathbf{1}000..000$

Last Activation?

sigmoid

(sequential)

## 4. Architecture ?

Sounds like it should be a RNN

## 5. Loss?

$$L = -(y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))$$

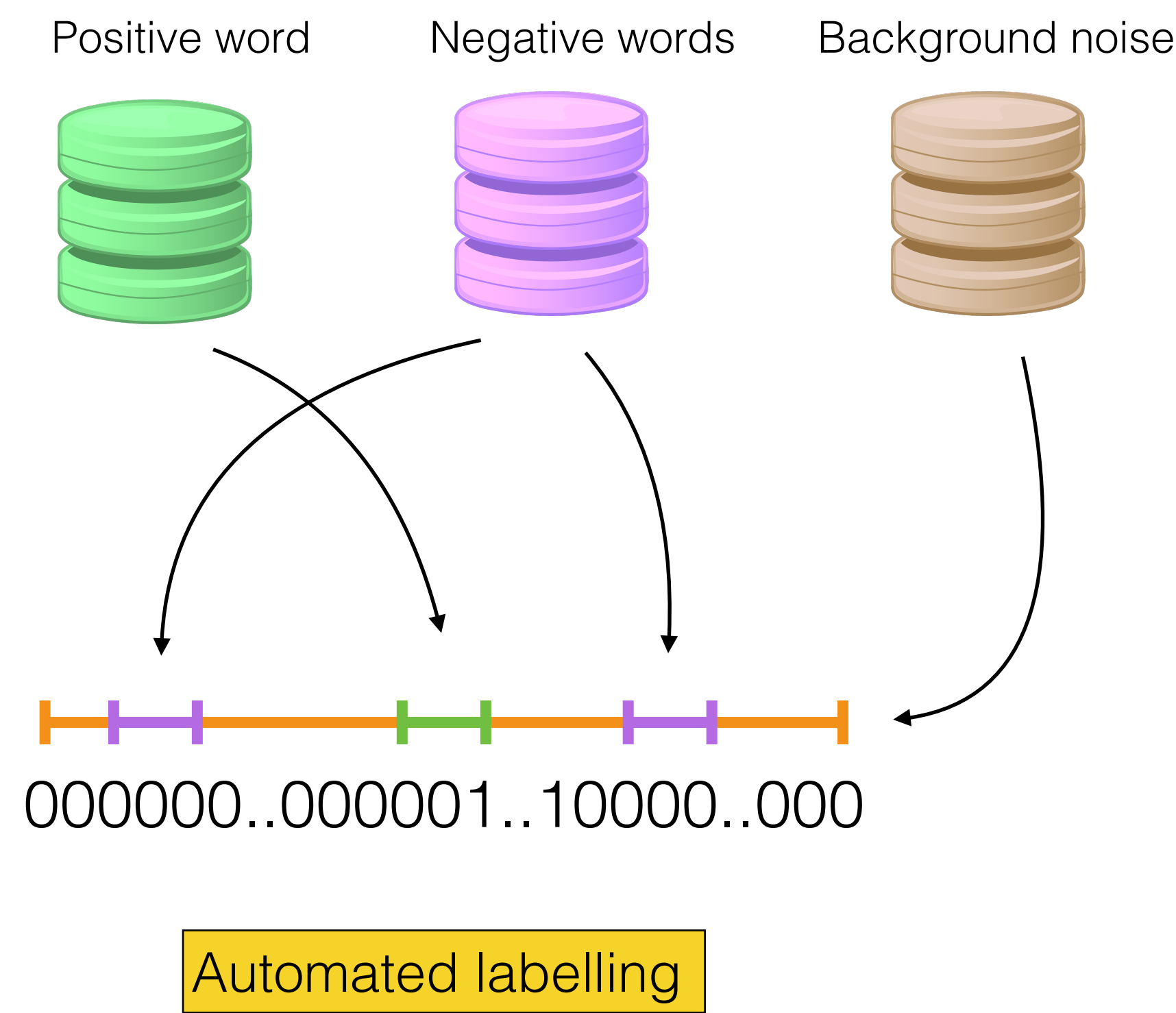
(sequential)



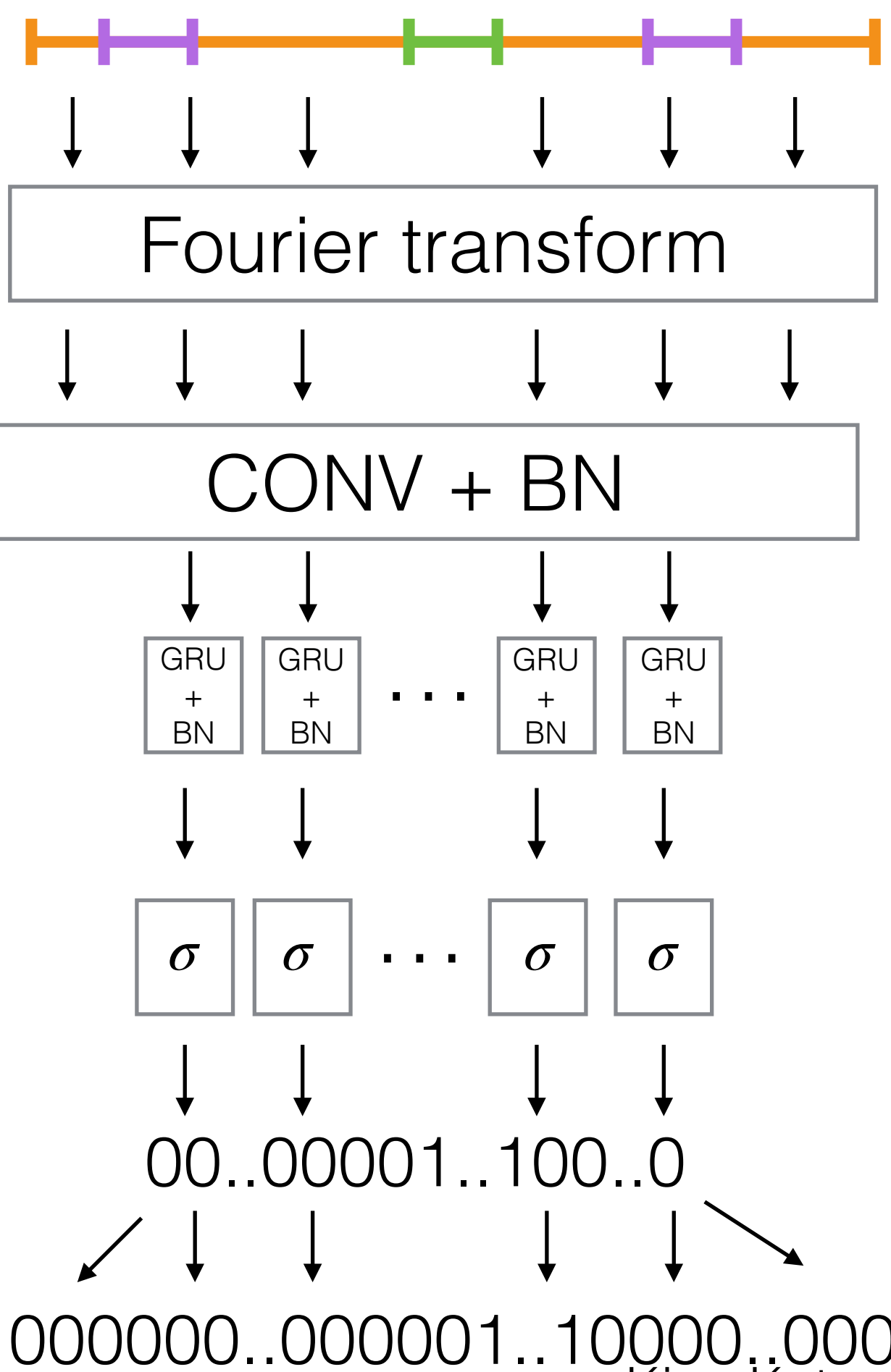
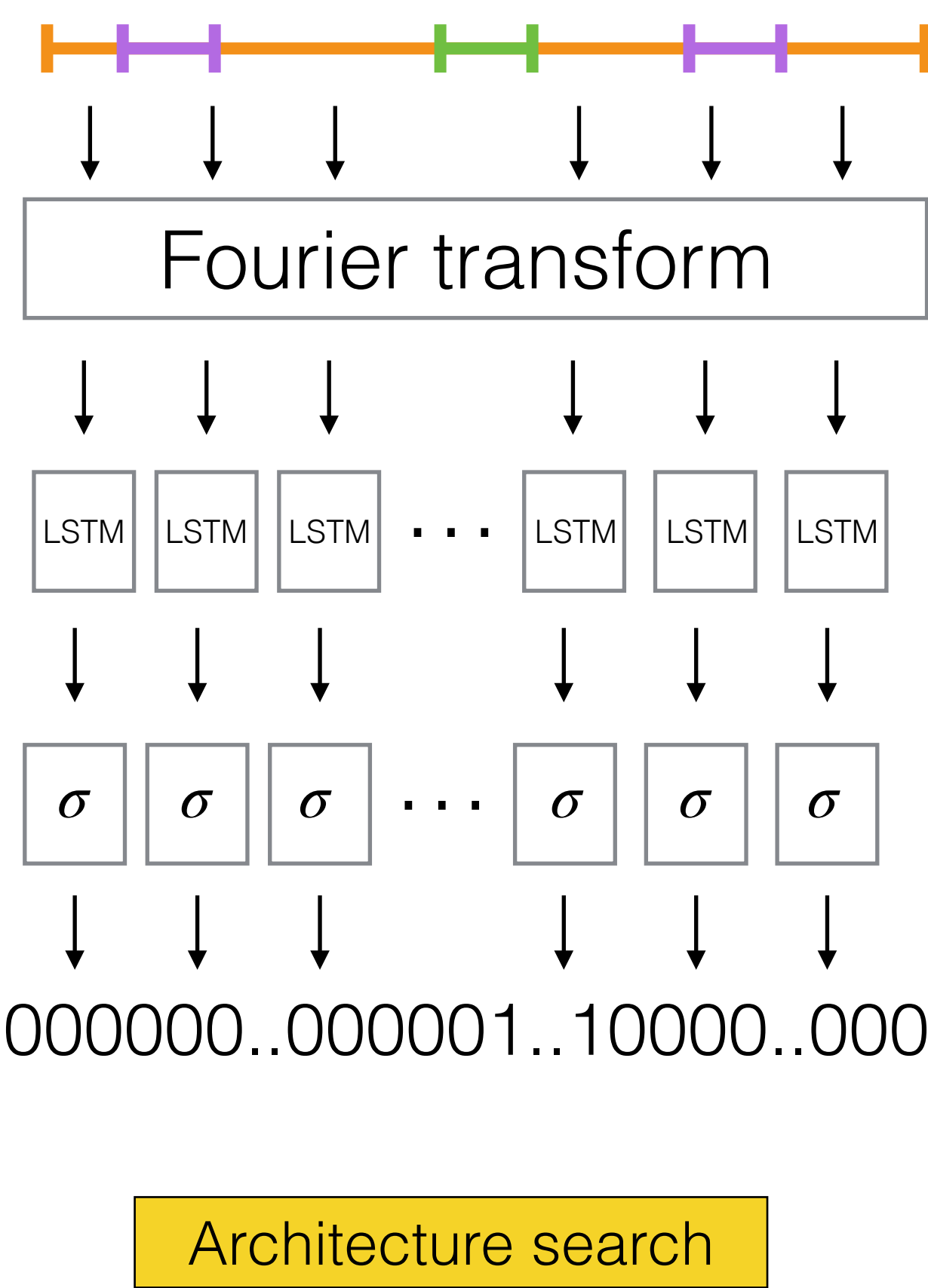
# Case study: Trigger word detection

What is critical to the success of this project?

## 1. Strategic data collection/ labelling process



## 2. Architecture search & Hyperparameter tuning



## Summary of learnings: Trigger word detection

- Your **data collection strategy** is critical to the success of your project. (If applicable) Don't hesitate to get out of the building.
- You can gain insights on your labelling strategy by using a **human experiment**.
- **Refer to expert advice** to earn time and be guided towards a good direction.

# Case Study: Face Verification

**Goal:** A school wants to use Face Verification for validating student IDs in facilities (dinning halls, gym, pool ...)

## 1. Data?

Picture of every student  
labelled with their name



Bertrand

## 2. Input?



Resolution?  
(412, 412, 3)

## 3. Output?

$y = 1$  (it's you)  
or  
 $y = 0$  (it's not you)



# Case Study: Face Verification

**Goal:** A school wants to use Face Verification for validating student IDs in facilities (dinning halls, gym, pool ...)

## 4. What architecture?

Simple solution:



database image

compute distance  
pixel per pixel



if less than threshold  
then  $y=1$



input image

Issues:

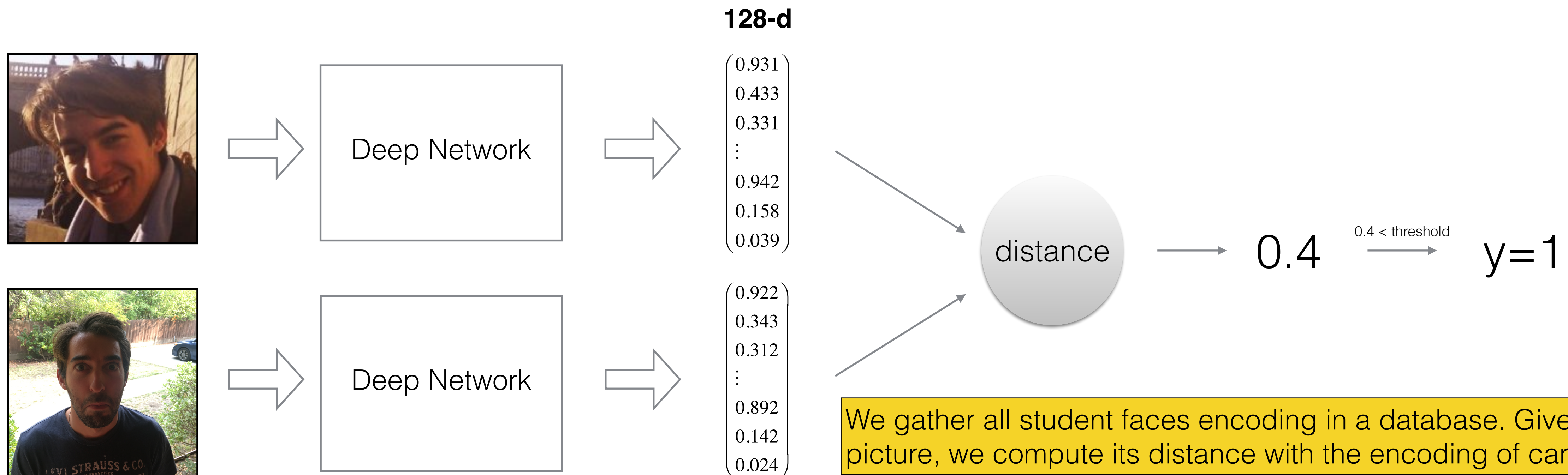
- Background lighting differences
- A person can wear make-up, grow a beard...
- ID photo can be outdated

# Case Study: Face Verification

**Goal:** A school wants to use Face Verification for validating student IDs in facilities (dinning halls, gym, pool ...)

## 4. What architecture?

Our solution: encode information about a picture in a vector





# Case Study: Face Verification

**Goal:** A school wants to use Face Verification for validating student IDs in facilities (dinning hall, gym, pool ...)

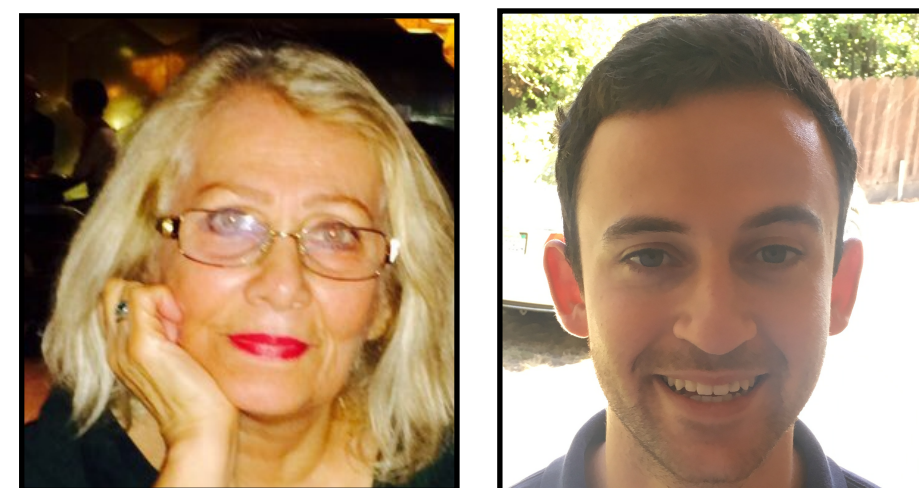
## 4. Loss? Training?

We need more data so that our model understands how to encode:  
Use public face datasets

What we really want:

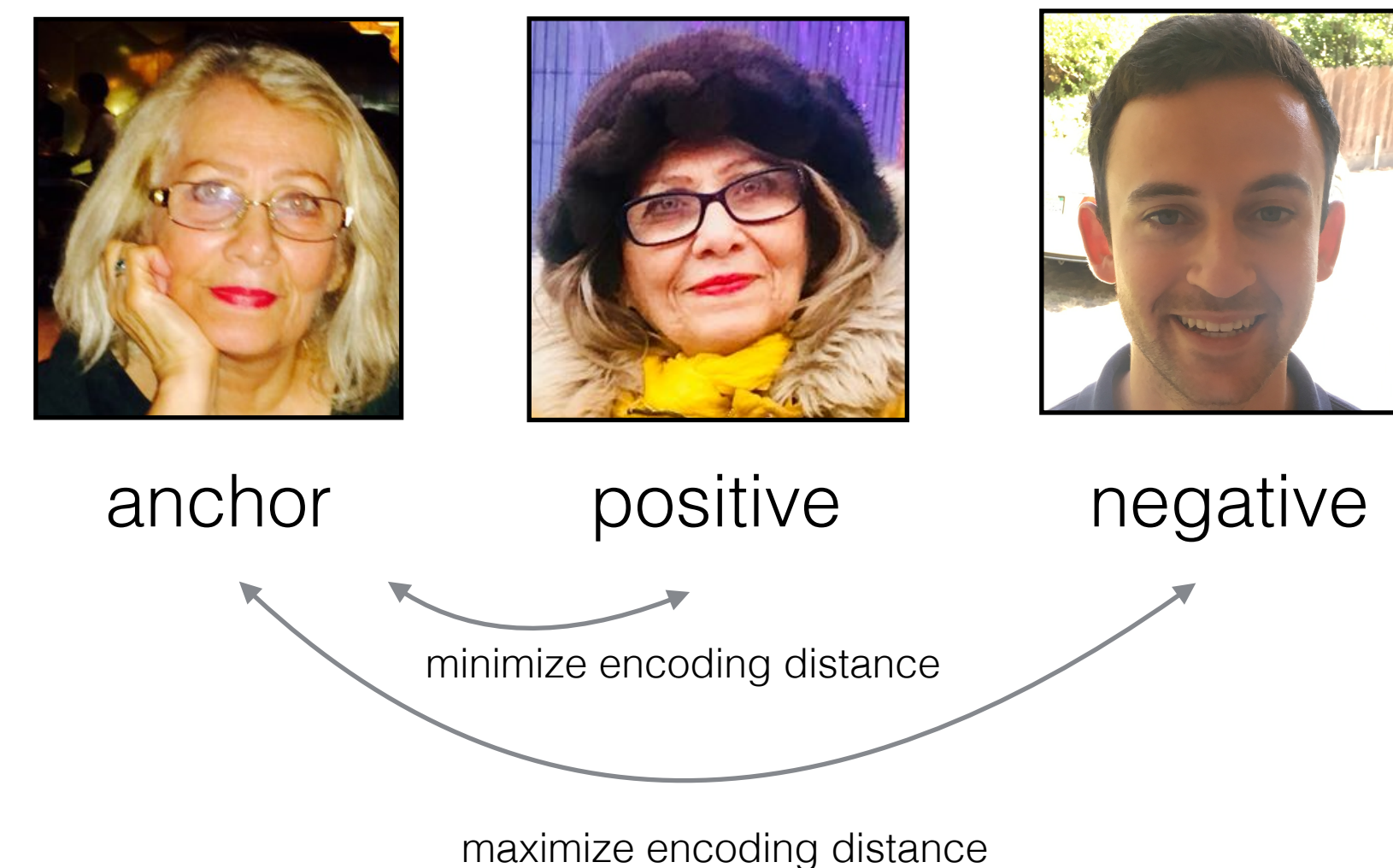


similar encoding



different encoding

So let's generate triplets:





# Case Study: Face Verification

What we really want:



similar encoding

different encoding

So let's generate triplets:



anchor

positive

negative

minimize encoding distance

maximize encoding distance

## Which loss should you minimize?

$$L = \left\| \text{Enc}(A) - \text{Enc}(P) \right\|_2^2$$

$$- \left\| \text{Enc}(A) - \text{Enc}(N) \right\|_2^2$$

A

$$L = \left\| \text{Enc}(A) - \text{Enc}(N) \right\|_2^2$$

$$- \left\| \text{Enc}(A) - \text{Enc}(P) \right\|_2^2$$

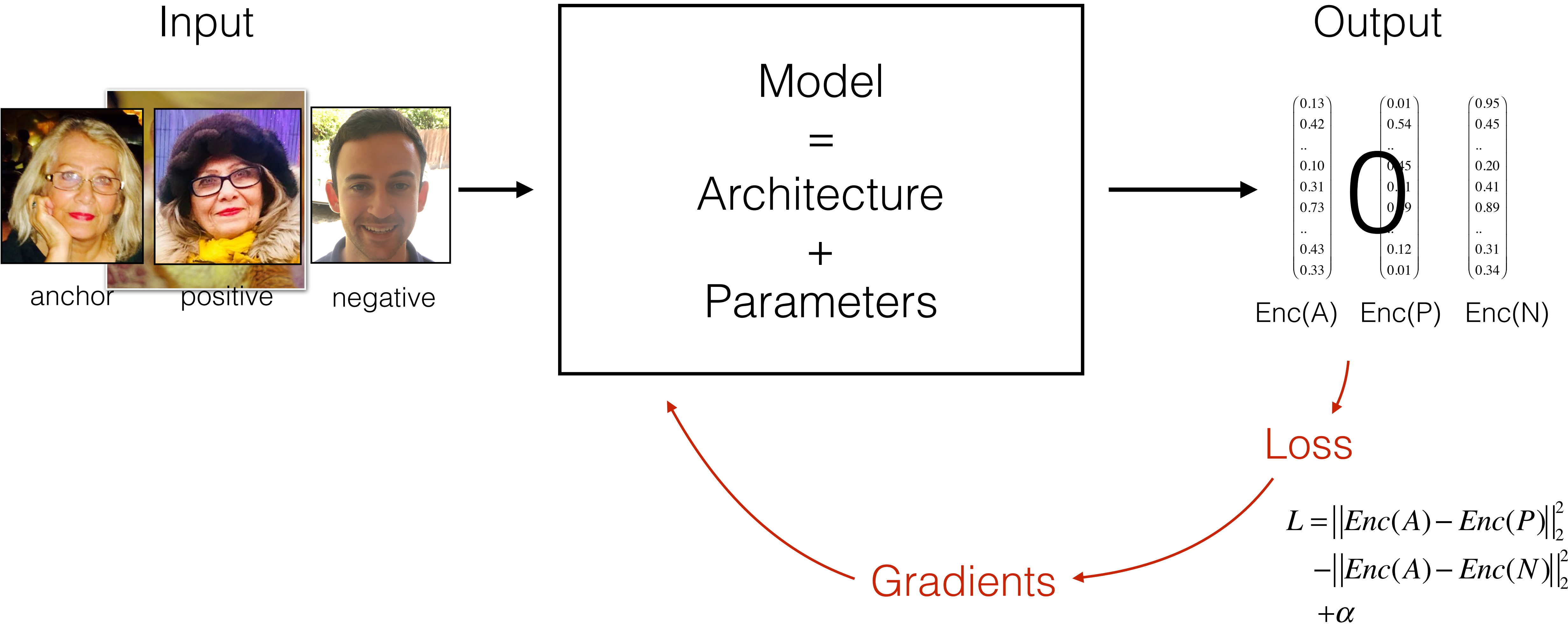
B

$$L = \left\| \text{Enc}(P) - \text{Enc}(N) \right\|_2^2$$

$$- \left\| \text{Enc}(P) - \text{Enc}(A) \right\|_2^2$$

C

# Case Study: Face Verification





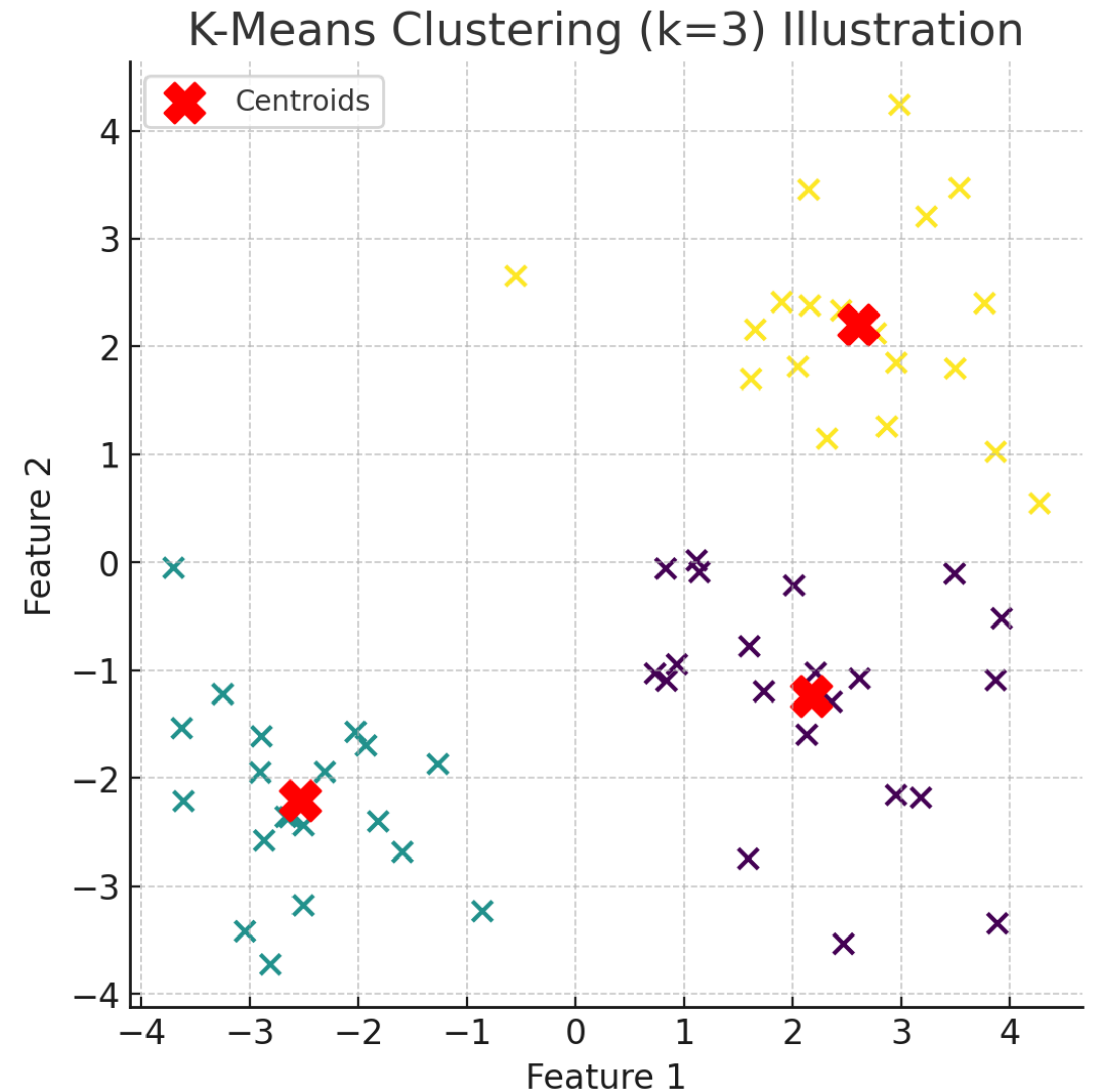
# Case Study: Face Identification and Face Clustering

**Goal:** A school wants to use Face Identification for recognize students in facilities (dinning hall, gym, pool ...)

K-Nearest Neighbors

**Goal:** You want to use Face Clustering to group pictures of the same people on your smartphone

K-Means Algorithm



# Summary of learnings: Face Verification

- In face verification, we have used an **encoder network** to learn a lower dimensional representation (called “**encoding**”) for a set of data by training the network to **focus on non-noisy signals**.
- **Triplet loss** is a loss function where an (**anchor**) input is compared to a **positive** input and a **negative** input. The distance from the anchor input to the positive input is minimized, whereas the distance from the anchor input to the negative input is maximized.
- You learned the difference between **face verification, face identification and face clustering**.

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# From Supervised to Self-Supervised Approaches

**Goal:** Labeling can be expensive. How would you create embeddings for facial images without any labels?

In **contrastive self-supervised learning**, you generate *paired views* from the same image (via augmentation) and force representations of those views to agree (positive pairs), while pushing apart other images (negative pairs).

The idea: if a model can reliably tell that two transformed views come from the same image, then the embedding it learned is meaningful (captures content invariant to those augmentations).

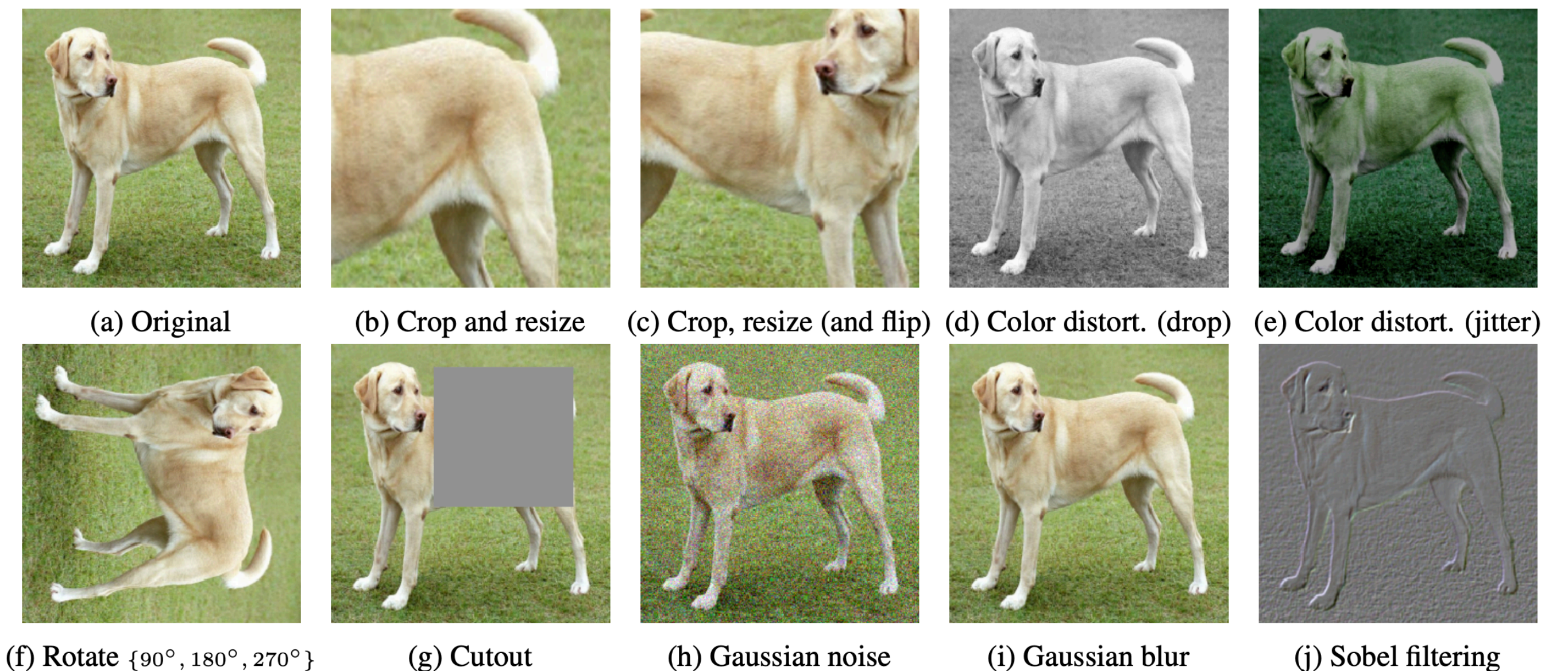


Figure 4. Illustrations of the studied data augmentation operators. Each augmentation can transform data stochastically with some internal parameters (e.g. rotation degree, noise level). Note that we *only* test these operators in ablation, the *augmentation policy* used to train our models only includes *random crop* (with *flip* and *resize*), *color distortion*, and *Gaussian blur*. (Original image cc-by: Von.grzanka)

This shift from supervised triplets to self-supervised pairs is why modern models can be trained on **billions of unlabeled images**. And the embeddings they learn power today's foundation models.

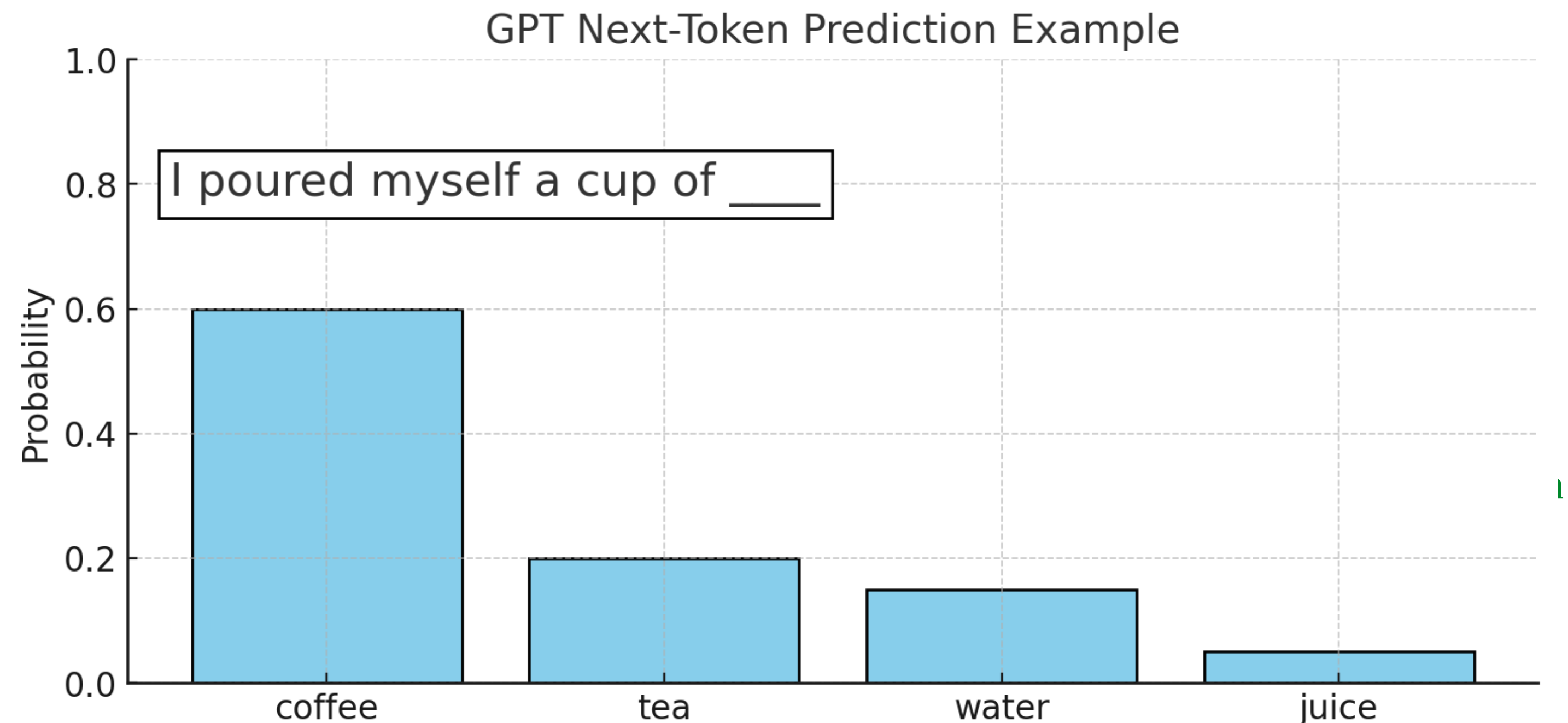


# GPT also uses self-supervision, but with text.

The principle is the same: predict what belongs together, and push away what doesn't. In images, SimCLR says: two augmented views of the same photo should be close. In text, GPT does says: the next word should be consistent with the context.”

What emergent behaviors stem from next-token prediction?

1. “I poured myself a cup of \_\_\_\_\_” → The model learns **event**
2. “The capital of France is \_\_\_\_\_” → Predicting the next token
3. “She unlocked her phone using her \_\_\_\_\_” → The model **learns meaning**
4. “The cat chased the \_\_\_\_\_” → Multiple completions are plausible
5. “If it’s raining, I should bring an \_\_\_\_\_” → The model can **reason about likely consequences**



**Emergent behaviors** are unexpected capabilities that arise from simple training objectives at scale, without being explicitly taught or labeled.

**Self-supervision isn't just about text. It's about creating a prediction task from raw data itself.**

- **Text** → *Next-token prediction (GPT)*
- **Images** → *Contrastive learning (SimCLR)*

**What other examples can you think of?**

- **Audio** → *Predict missing audio chunks from surrounding sound.*
- **Video** → *Predict the next frame, or the correct order of shuffled frames.*
- **Biology / Proteins** → *Hide one amino acid in a sequence and predict it*

**But the world is multimodal. We experience words, images, sounds, and actions together. How do we connect them?**

The multimodal embedding task is technically supervised (because you rely on *paired* data like “image  $\leftrightarrow$  caption”), but it’s a very *weak* or *naturally occurring* form of supervision, not like ImageNet-style hand labeling.



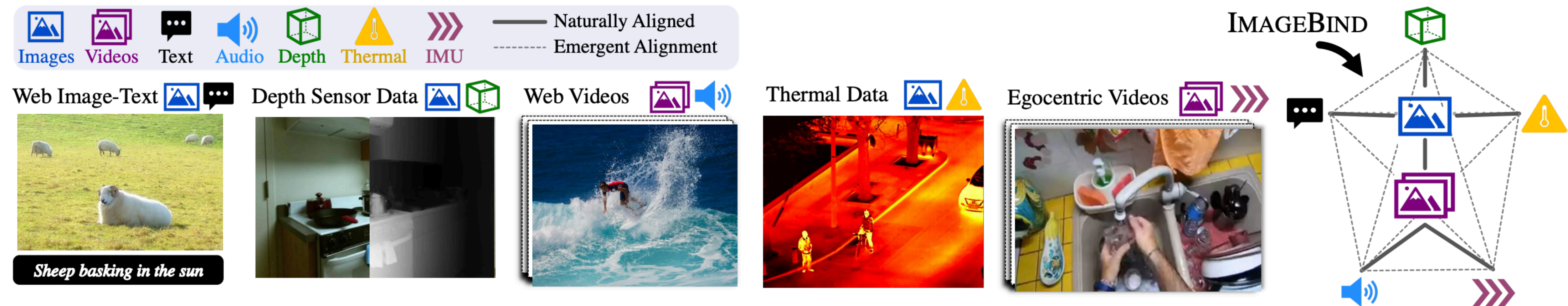
The cat is looking at the camera, a bit confused

This is often called “**weakly supervised**”.

Can you think of some examples where different types of data naturally come paired together?



# How would you align modalities that are not naturally occurring together in available datasets?



**Figure 2. IMAGEBIND overview.** Different modalities occur naturally aligned in different data sources, for instance images+text and video+audio in web data, depth or thermal information with images, IMU data in videos captured with egocentric cameras, *etc.* IMAGE-BIND links all these modalities in a common embedding space, enabling new emergent alignments and capabilities.

# Summary of learnings: Embeddings (Self-Supervised Learning)

- **Embedding**: An encoding where closeness = similarity.
- **Self-supervised learning** is a method where models create their own training labels from raw data by solving prediction tasks inherent in the data itself.
- **Contrastive learning** uses **data transformations** to create similar pairs, pulling them together while pushing others apart.
- **Next-token prediction** trains a model to guess the most likely next word in a sequence given all the previous words.
- **Weakly supervised** = training models with imperfect or automatically generated labels instead of precise human ones
- We can learn a **shared embedding space** across modalities (images, text, audio, depth, thermal, and IMU sensors) using text as the **central pivot**.